Group

Electromagnetism 2: Joule heat virtual lab

Full name: _____ Nickname: _____ ID (short): _____ Score: _____/5

Group members:

1. Nickname: _____ ID: ___ Score: ____/5
2. Nickname: _____ ID: ___ Score: ____/5
3. Nickname: _____ ID: ___ Score: ____/5

Learning topics: Joule heat

Scoring criteria:

- Deduction of 1 to 5 points for each incorrect element (significant figures, graph plotting, calculations and conclusion).
- Deduction of 20 points for late submissions.
- Deduction of 5 to 20 points for low-quality work.
- 5 out of 100 points will be based on group members evaluation (not including yourself).
- 5 out of 100 points will be based on teacher's evaluation.

Rubric score for group members evaluation:

5	4	3	2	1
Proactively collaborates, encourages diverse perspectives, and works well with all team members.	Collaborates effectively but may occasionally struggle to integrate different viewpoints.	Collaborates adequately, but occasionally resists input from others or works independently.	Collaboration is limited, and there is difficulty in adapting to the team's needs.	Does not collaborate, undermining the team's ability to function effectively.

Experimental report

1. **Part 1: Unknown wire measurement** [10 points]
2. **Part 2: Heating water in isolated system**
    (a) **Data recording** [15 points]
    (b) **Data analysis** [15 points]
    (c) **Discussion and conclusion** [15 points]
3. **Part 3: Heating ethanol in isolated system**
    (a) **Data recording** [15 points]
    (b) **Data analysis** [15 points]
    (c) **Discussion and conclusion** [15 points]
4. **Part 4 (optional): Heating water in closed system**
    (a) **Data recording** [0 points]
    (b) **Data analysis** [0 points]
    (c) **Discussion and conclusion** [0 points]